Advanced AI Medical Intelligence Platform

Project Report

An end-to-end academic system for medical image classification, explainable AI, LLM-assisted reporting, REST APIs, database-backed history, and web deployment.

Objective

Build a complete AI application capable of analyzing medical images, predicting disease classes with deep learning, explaining predictions through Grad-CAM, generating cautious AI-assisted reports, exposing REST APIs, storing prediction history, and providing a usable UI.

Architecture

The platform separates concerns across FastAPI routes, ML inference, Grad-CAM explainability, report generation, SQLAlchemy persistence, and Streamlit presentation. This keeps the project maintainable and ready for deployment.

Deep Learning

The model is a compact convolutional neural network with batch normalization, pooling, dropout, and a linear classification head. The included training script creates reproducible educational weights and can be adapted to NIH ChestX-ray14, CheXpert, MIMIC-CXR, or RSNA datasets.

Explainable AI

Grad-CAM computes gradients for the predicted class and overlays the resulting heatmap on the original image, helping reviewers inspect which regions influenced the model.

LLM Integration

The report generator supports optional OpenAI-backed draft reports and falls back to a deterministic safe report when no API key is configured. Every report includes non-diagnostic safety language.

Deployment

The repository includes requirements.txt, Dockerfile, docker-compose.yml, and clear run commands for local or cloud deployment.

API Endpoints

Endpoint	Purpose
GET /api/v1/health	Service health check
POST /api/v1/predict	Image prediction, probabilities, report, and Grad-CAM URL
GET /api/v1/history	Recent prediction records
GET /api/v1/heatmap/{filename}	Generated explainability overlay

Safety and Limitations

This project is not a medical device. It is an academic decision-support demonstration. Clinical deployment would require licensed data, external validation, privacy controls, security review, monitoring, and regulatory assessment.

Deliverables

Complete source code, model artifact path, reproducible training script, README, PDF report, requirements.txt, Dockerfile, database design, REST APIs, Streamlit UI, and deployment guidance are included.